Chapter 11. Analyzing Repeated Categorical Response Data

Marginal models, GEE, Markov chains

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Many studies observe a categorical response repeatedly for each subject: at several times (longitudinal studies), under several conditions (crossover trials), or for several units of a *cluster* (teeth in a mouth, fetuses in a litter, children in a family). Observations within a cluster tend to be more alike than observations from different clusters, and analyses ignoring this can be badly wrong. This chapter generalizes Chapter 10 from pairs to sets of T responses and adds explanatory variables. Section 11.1 compares the T margins; Section 11.2 fits marginal models by ML; Sections 11.3–11.4 develop the computationally simpler *generalized estimating equations* (GEE), a multivariate quasi-likelihood; Section 11.5 models each response given the previous ones (transitional models).

11.1 Comparing marginal distributions: multiple responses

Motivation. Usually the T marginal distributions (e.g. the success probability at each week of a treatment) matter more than the full joint dependence. We extend the marginal analyses of Chapter 10 from $T = 2$ to $T > 2$.

11.1.1 Binary marginal models and marginal homogeneity

For T binary responses $(Y_1, \ldots, Y_T)$ the marginal logit model

$$\operatorname{logit} P(Y_t = 1) = \alpha + \beta_t, \qquad t = 1, \ldots, T, \tag{11.1}$$

with a constraint such as $\beta_T = 0$, extends the matched-pairs model of Section 10.2.1. For a sequence $\mathbf{i} = (i_1, \ldots, i_T)$, $i_t \in \{0, 1\}$, let $\pi_{\mathbf{i}} = P(Y_1 = i_1, \ldots, Y_T = i_T)$; these are the cell probabilities of the 2^T table. The sample cell proportions are the ML estimates of $\{\pi_{\mathbf{i}}\}$, and the sample proportion with $y_t = 1$ is the ML estimate of $P(Y_t = 1)$. Model (11.1) is saturated; **marginal homogeneity** $P(Y_1 = 1) = \cdots = P(Y_T = 1)$ is the special case $\beta_1 = \cdots = \beta_T$. Even this one-parameter model is awkward to fit by ML, because the multinomial likelihood involves the 2^T joint probabilities rather than the T margins.

With $n_{\mathbf{i}}$ the count in cell $\mathbf{i}$, $p_{\mathbf{i}} = n_{\mathbf{i}}/n$, and $\hat{\pi}^{MH}$ the ML fit under marginal homogeneity, the likelihood-ratio statistic is

$$-2\big[L(\hat{\pi}^{MH}) - L(\mathbf{p})\big] = 2\sum_{\mathbf{i}} n_{\mathbf{i}} \log\left(p_{\mathbf{i}}/\hat{\pi}^{MH}_{\mathbf{i}}\right), \tag{11.2}$$

asymptotically χ^2_{T-1} under H_0, since (11.1) has $T-1$ more parameters than marginal homogeneity.

11.1.2 Crossover drug comparison example

Example 11.1 (Crossover study of three drugs)

Each of 46 subjects with a chronic condition used drugs A, B, C at three times; the reaction was favorable (F) or unfavorable (U). Assume no carryover effects and a stable condition.

	A favorable		A unfavorable	
	B F	B U	B F	B U
C favorable	6	2	2	6
C unfavorable	16	4	4	6

The proportions favorable are $(0.61, 0.61, 0.35)$ for (A, B, C). Statistic (11.2) equals 5.95 (df $= 2$, $P = 0.05$). For Bonferroni intervals with overall error ≤ 0.05, each of the three pairwise intervals uses confidence level $1 - 0.05/3$, so $z = 2.39$. For A vs C, $0.609 - 0.348 = 0.261$ with

SE 0.108 (Theorem 10.1), giving $0.261 \pm 2.39(0.108) = (0.002, 0.520)$; the same holds for B vs C. Exact binomial tests on the discordant pairs of each 2×2 table give $P = 1.0$ (A vs B) and $P = 0.036$ (A vs C, B vs C). The sample is small, so these conclusions are tentative.

11.1.3 Modeling margins of a multicategory response

For an I-category response, the saturated baseline-category model is

$$\log \frac{P(Y_t = j)}{P(Y_t = I)} = \beta_{tj}, \qquad t = 1, \ldots, T, \quad j = 1, \ldots, I-1,$$

and marginal homogeneity is $\beta_{1j} = \cdots = \beta_{Tj}$ for all j; its likelihood-ratio test has the form (11.2) with $\mathrm{df} = (T-1)(I-1)$. For an ordinal response, a model between these two is

$$\operatorname{logit} P(Y_t \le j) = \alpha_j + \beta_t, \qquad t = 1, \ldots, T, \quad j = 1, \ldots, I-1, \tag{11.3}$$

with $\beta_T = 0$ and $\alpha_1 < \cdots < \alpha_{I-1}$; marginal homogeneity is $\beta_1 = \cdots = \beta_T$, a test with $\mathrm{df} = T - 1$.

11.1.4 Wald and generalized CMH score tests of marginal homogeneity

Let $p_j(t)$ be the sample proportion in category j at occasion t, $\bar{p}_j = \sum_t p_j(t)/T$, $d_j(t) = p_j(t) - \bar{p}_j$, and $\mathbf{d}$ the vector of $d_j(t)$, $t \le T-1$, $j \le I-1$. With $\hat{\mathbf{V}}$ the estimated covariance matrix of $\sqrt{n}\,\mathbf{d}$, Bhapkar's Wald statistic

$$W = n\,\mathbf{d}'\hat{\mathbf{V}}^{-1}\mathbf{d} \tag{11.4}$$

is asymptotically $\chi^2_{(I-1)(T-1)}$ under marginal homogeneity, generalizing Proposition 10.14.

Other tests are generalized CMH statistics. Display the data as n strata, one per subject; stratum k is a $T \times I$ table whose row t has a 1 in the column of the outcome at occasion t. Conditional independence of occasion and response given subject means that, for each subject, the T responses are exchangeable; this implies marginal homogeneity, and the generalized CMH test of conditional independence is a test of marginal homogeneity whose null distribution is generated by this stronger exchangeability (Darroch 1981). For $I = T = 2$ it is McNemar's test (Theorem 10.5); for $I = 2$, $T > 2$ it is Cochran's Q.

Theorem 11.2 (Cochran's Q as a generalized CMH statistic)

For binary responses, let R_k be the number of successes of subject k, C_t the number of successes at occasion t and $N = \sum_t C_t = \sum_k R_k$. The generalized CMH statistic (general association) for the $T \times 2 \times n$ subject-stratified table is

$$Q = \frac{T(T-1)\sum_{t=1}^{T}(C_t - N/T)^2}{TN - \sum_k R_k^2},$$

which is asymptotically χ^2_{T-1} under marginal homogeneity. For $T = 2$, $Q = (n_{21} - n_{12})^2/(n_{21} + n_{12})$.

Proof. [Supplement]

Step 1. Null distribution. Under conditional independence with fixed stratum margins, the R_k successes of subject k are allocated to the T occasions completely at random (each of the $\binom{T}{R_k}$ allocations equally likely), independently over subjects. Since column 2 is determined by column 1, the CMH statistic is the quadratic form in $\mathbf{C} = (C_1, \ldots, C_T)'$ about its null mean, with a generalized inverse of its null covariance.

Step 2. Moments for one subject. Let Y_{kt} indicate success at occasion t and $c_k = (R_k/T)(1-R_k/T)$. Under random allocation (sampling without replacement),

$$E(Y_{kt}) = \frac{R_k}{T}, \qquad \mathrm{Var}(Y_{kt}) = c_k, \qquad \mathrm{Cov}(Y_{kt}, Y_{ks}) = -\frac{c_k}{T-1} \ (t \neq s),$$

i.e. $\mathrm{Cov}(\mathbf{Y}_k) = \frac{T}{T-1} c_k (\mathbf{I} - \frac{1}{T}\mathbf{J})$ with $\mathbf{J}$ the matrix of ones.

Step 3. Summing over independent subjects, $E(\mathbf{C}) = (N/T)\mathbf{1}$ and

$$\mathrm{Cov}(\mathbf{C}) = \kappa(\mathbf{I} - \tfrac{1}{T}\mathbf{J}), \qquad \kappa = \frac{T}{T-1}\sum_k c_k = \frac{\sum_k R_k(T-R_k)}{T(T-1)} = \frac{TN - \sum_k R_k^2}{T(T-1)}.$$

Step 4. $\mathbf{P} = \mathbf{I} - \frac{1}{T}\mathbf{J}$ is the projection onto $\{\mathbf{v} : \mathbf{1}'\mathbf{v} = 0\}$, so $\mathbf{P}$ is its own generalized inverse and $\mathbf{C} - E\mathbf{C}$ lies in its range. The quadratic form is

$$(\mathbf{C} - E\mathbf{C})'\frac{\mathbf{P}}{\kappa}(\mathbf{C} - E\mathbf{C}) = \frac{1}{\kappa}\sum_t \left(C_t - N/T\right)^2 = Q.$$

The rank of $\mathbf{P}$ is $T-1$, which gives the χ^2_{T-1} limit (CLT for the sum of independent subject vectors).

Step 5. $T = 2$: only discordant subjects ($R_k = 1$) contribute to the denominator, which equals $2N - \sum R_k^2 = n_{12} + n_{21}$ after cancelling concordant contributions ($R_k = 0$ gives 0, $R_k = 2$ gives $2 \cdot 2 - 4 = 0$). Also $C_1 - C_2 = n_{12} - n_{21}$, so $\sum_t (C_t - N/2)^2 = (n_{12} - n_{21})^2/2$ and $Q = 2 \cdot \frac{1}{2}(n_{12} - n_{21})^2/(n_{12} + n_{21})$. ■

For an ordinal response with fixed scores, the generalized CMH statistic for variation among the T occasion means is the appropriate single-df version.

11.2 Marginal modeling: maximum likelihood approach

Motivation. Usually we want to know how the margins depend on explanatory variables (treatment, severity, time). Marginal logit models answer this, and ML fitting of them is possible by maximizing the full multinomial likelihood subject to constraints on the margins.

11.2.1 Longitudinal mental depression example

Example 11.3 (Mental depression)

Subjects with mild or severe initial depression were randomized to a standard or a new drug; after 1, 2 and 4 weeks each was classified as normal (N) or abnormal (A):

Diagnosis	Treatment	NNN	NNA	NAN	NAA	ANN	ANA	AAN	AAA
Mild	Standard	16	13	9	3	14	4	15	6
Mild	New drug	31	0	6	0	22	2	9	0
Severe	Standard	2	2	8	9	9	15	27	28
Severe	New drug	7	2	5	2	31	5	32	6

Let $s = 1$ for severe, $d = 1$ for the new drug and time $t = 0, 1, 2$ ($\log_2$ of the week number, since the logit is often linear in log time). Sample proportions normal:

Diagnosis	Treatment	Week 1	Week 2	Week 4
Mild	Standard	0.51	0.59	0.68
Mild	New drug	0.53	0.79	0.97
Severe	Standard	0.21	0.28	0.46
Severe	New drug	0.18	0.50	0.83

The proportion increases with time in every group, faster for the new drug, and is higher for mild diagnosis. Sampling is multinomial over the 8 sequences independently in the 4 groups (32 cells), while a marginal model concerns the 12 marginal probabilities. The main-effects model

$$\text{logit}\, P(Y_t = 1) = \alpha + \beta_1 s + \beta_2 d + \beta_3 t$$

uses 4 parameters for 12 marginal logits (df = 8) and fits poorly, $G^2 = 34.6$: it forces a common time slope. With drug-by-time interaction,

$$\text{logit}\, P(Y_t = 1) = \alpha + \beta_1 s + \beta_2 d + \beta_3 t + \beta_4 dt,$$

$G^2 = 4.2$ (df = 7). The time slope is $\hat{\beta}_3 = 0.48$ (SE 0.12) for the standard drug and $\hat{\beta}_3 + \hat{\beta}_4 = 1.49$ (SE 0.14) for the new drug; $\hat{\beta}_4 = 1.01$ (SE 0.18), and $34.6 - 4.2 = 30.4$ is the LR test of $\beta_4 = 0$. Also $\hat{\beta}_1 = -1.29$ (SE 0.14): severe diagnosis multiplies the odds of normal response by $e^{-1.29} = 0.27$. $\hat{\beta}_2 = -0.06$ (SE 0.22) shows no drug difference at week 1; at time t the new-to-standard odds ratio is $e^{-0.06+1.01t}$.

11.2.2 Modeling a repeated multinomial response

For an I-category response, observation t has $I - 1$ marginal logits $\text{logit}_j(t)$ (baseline-category for nominal, cumulative logit $P(Y_t \le j)$ for ordinal responses), and a model has the form

$$\text{logit}_j(t) = \alpha_j + \boldsymbol{\beta}_j' \mathbf{x}_t, \qquad j = 1, \ldots, I-1.$$

For cumulative logits $\boldsymbol{\beta}_j$ may simplify to $\boldsymbol{\beta}$ (proportional odds). Some components of $\mathbf{x}_t$ may refer to the occasion t itself. Checking interaction is essential: are the effects of $\mathbf{x}$ the same at each t?

11.2.3 Insomnia example

Example 11.4 (Insomnia clinical trial)

In a randomized double-blind trial, patients reported time to fall asleep (minutes) before and after two weeks of an active hypnotic or a placebo:

		Follow-up			
Treatment	Initial	< 20	20–30	30–60	> 60
Active	< 20	7	4	1	0
	20–30	11	5	2	2
	30–60	13	23	3	1
	> 60	9	17	13	8
Placebo	< 20	7	4	2	1
	20–30	14	5	1	0
	30–60	6	9	18	2
	> 60	4	11	14	22

Sample marginal distributions:

Treatment	Occasion	< 20	20–30	30–60	> 60
Active	Initial	0.101	0.168	0.336	0.395
	Follow-up	0.336	0.412	0.160	0.092
Placebo	Initial	0.117	0.167	0.292	0.425
	Follow-up	0.258	0.242	0.292	0.208

Both distributions shift down, more for the active drug. With $t = 0$ (initial), 1 (follow-up) and $x = 1$ (active), 0 (placebo), the proportional odds model

$$\text{logit}\, P(Y_t \le j) = \alpha_j + \beta_1 t + \beta_2 x + \beta_3 tx \tag{11.5}$$

fitted by ML has $G^2 = 8.0$ (df $= 6$; 12 marginal logits, 6 parameters), with $\hat\beta_1 = 1.074$ (SE 0.162), $\hat\beta_2 = 0.046$ (SE 0.236), $\hat\beta_3 = 0.662$ (SE 0.244): evidence of interaction. Initially the active-vs-placebo cumulative odds ratio is $e^{0.046} = 1.04$; at follow-up it is $e^{0.046+0.662} = 2.03$. With scores $(10, 25, 45, 75)$ the initial means are 50.0 (active) and 50.3 (placebo); the mean decreases are 22.2 and 13.0, a difference of 9.2 (SE 3.0).

11.2.4 Comparisons that control for initial response

Marginal models do not always tell the whole story

If the treatment groups have identical initial marginal distributions (true up to sampling error under randomization) and identical conditional distributions of the follow-up response given the initial response, then by the law of total probability their follow-up margins are identical too. But if the initial margins differ, the changes in the margins can differ between groups even though the conditional follow-up distributions are identical. It is then more informative to compare follow-up responses controlling for the initial response y_1:

$$\text{logit}\, P(Y_2 \le j) = \alpha_j + \beta_1 x + \beta_2 y_1, \tag{11.6}$$

an ordinal analogue of analysis of covariance and a simple *transitional* model (Section 11.5).

11.3 Marginal modeling: generalized estimating equations (GEE) approach

Motivation. ML marginal fitting becomes impractical when T is large or there are many (possibly continuous) predictors, and there is no simple multivariate distribution for categorical responses. GEE specifies only the marginal mean, a variance function and a working guess at the correlation, and still estimates the regression parameters consistently.

11.3.1 Generalized estimating equation methodology: basic ideas

For a multivariate response $(Y_1, \ldots, Y_T)$ (T may vary by subject), the quasi-likelihood idea of Section 4.7 is applied to each margin: a model links $\mu_t = E(Y_t)$ to a linear predictor, and a variance function $v(\mu_t)$ describes $\text{Var}(Y_t)$. In addition one chooses a *working correlation* structure for $\{Y_t\}$. The estimates solve multivariate quasi-likelihood equations, the **generalized estimating equations** (Liang and Zeger 1986, building on econometric work of Gourieroux–Monfort–Trognon, Hansen and White).

Definition (Working correlation structures)

For cluster i with T_i observations, a working correlation matrix $\mathbf{R}(\boldsymbol{\alpha})$ specifies $\text{Corr}(Y_{it}, Y_{is})$:
(i) *independence*: $\mathbf{R} = \mathbf{I}$;
(ii) *exchangeable*: $\text{Corr}(Y_{it}, Y_{is}) = \alpha$ for all $t \neq s$;
(iii) *autoregressive* AR(1): $\text{Corr}(Y_{it}, Y_{is}) = \alpha^{|t-s|}$;
(iv) *unstructured*: a separate α_{ts} for each pair.

The key properties, proved in Section 11.4:

(a) If the mean model (link and linear predictor) is correct, the GEE estimators are consistent *even if the working correlation is wrong*: consistency depends on the first moment, not the second.

(b) The naive (model-based) standard errors are wrong when the working correlation is wrong; GEE uses the empirical dependence in the data to compute *robust* (sandwich) standard errors.

Even the naive independence structure gives good estimates, with corrected standard errors. A good working correlation improves efficiency; estimators under independence can be surprisingly efficient when correlations are weak to moderate. One can check sensitivity by comparing several structures; when they differ substantially, the correlation structure deserves study. Exchangeable is a sensible default (one extra parameter); unstructured is flexible but costs efficiency for large T.

Limitations of GEE

GEE does not specify a joint distribution, so there is no likelihood: no likelihood-ratio tests of fit or model comparison. Inference uses Wald statistics with the sandwich covariance. Unless the number of clusters is large, sandwich SEs tend to underestimate the true ones and are themselves variable; score-type tests based on the quasi-likelihood (Boos 1992; Rotnitzky and Jewell 1990) may be more trustworthy. Some critics object to the lack of likelihood; others view GEE as an estimation method rather than a model.

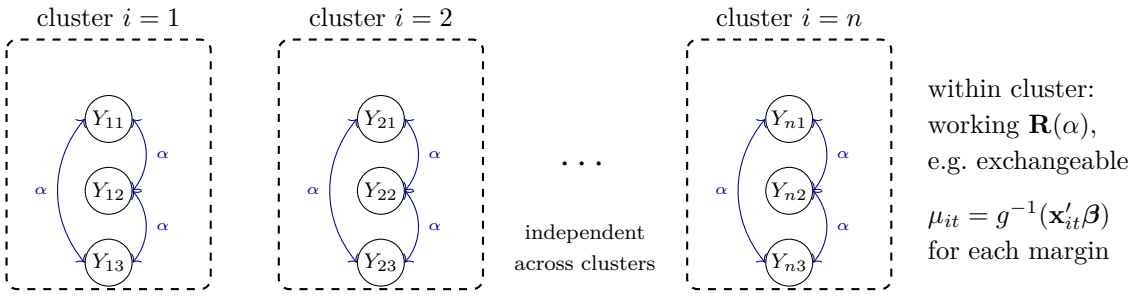


Figure 11.1. Clustered data: n independent clusters, dependent observations within a cluster. GEE models each margin and uses a working correlation within clusters; asymptotics are in the number of clusters n.

11.3.2 Longitudinal mental depression example

Example 11.5 (Depression: GEE fit)

For Example 11.3 with the interaction model, GEE gives results close to ML for every working correlation. With the exchangeable structure, $\hat{\beta}_3 = 0.48$ (SE 0.12) and $\hat{\beta}_4 = 1.02$ (SE 0.19). With the independence structure:

Parameter	Estimate	Naive SE (independence)	Robust (sandwich) SE
Intercept	-0.028	0.164	0.174
Diagnosis (severe)	-1.314	0.146	0.146
Drug (new)	-0.060	0.222	0.229
Time	0.482	0.115	0.120
Drug $\times$ time	1.017	0.189	0.188

The estimates and naive SEs are those of ordinary logistic regression on $3 \times 340 = 1020$ observations treated as independent; the robust SEs use the empirical within-subject dependence. The exchangeable estimate of the common correlation is $\hat{\alpha} = -0.003$: the three responses look pairwise uncorrelated, unusual for repeated measures, which is why all analyses agree so closely.

11.3.3 GEE approach for multinomial responses: insomnia example

GEE extends to marginal models for multinomial responses (Lipsitz, Kim and Zhao 1994): for each pair of outcome categories one chooses a working correlation for pairs of occasions, and each multinomial observation uses its $(I-1) \times (I-1)$ multinomial covariance.

Example 11.6 (Insomnia: GEE fit)

For model (11.5) with independence working correlation, GEE gives $\hat{\beta}_1 = 1.038$ (SE 0.168), $\hat{\beta}_2 = 0.034$ (SE 0.238), $\hat{\beta}_3 = 0.708$ (SE 0.244), similar to ML: time to fall asleep decreased more for the active treatment.

11.4 Quasi-likelihood and its GEE multivariate extension: details

Motivation. Why is GEE consistent when the correlation is wrong, and where does the sandwich come from? Both follow from one general fact about estimating equations with mean-zero estimating functions. We prove it once (Lemma 11.8) and apply it to quasi-likelihood and to GEE.

For subject i, $i = 1, \ldots, n$, let y_i have mean μ_i, variance function $v(\mu_i)$, and linear predictor $\eta_i = g(\mu_i) = \sum_j \beta_j x_{ij} = \mathbf{x}_i'\boldsymbol{\beta}$. The **quasi-likelihood** (QL) estimates solve the quasi-score equations

$$\mathbf{u}(\boldsymbol{\beta}) = \sum_{i=1}^{n} \Big(\frac{\partial \mu_i}{\partial \boldsymbol{\beta}}\Big) \frac{y_i - \mu_i}{v(\mu_i)} = \mathbf{0}, \qquad \mu_i = g^{-1}(\mathbf{x}_i'\boldsymbol{\beta}). \tag{11.7}$$

Proposition 11.7 (Motivations for the quasi-score equations)

(a) If y_i has a natural exponential family distribution with variance function v, (11.7) are the GLM likelihood equations.
(b) With $v(\mu_i)$ replaced by known constants v_i, (11.7) are the normal equations for minimizing $\sum_i (y_i - \mu_i)^2 / v_i$.
(c) Since $\partial \mu_i / \partial \beta_j = (\partial \mu_i / \partial \eta_i) x_{ij}$, (11.7) depend only on the link and the variance function.

Proof. [Supplement] (a) For $f(y;\theta) = \exp\{y\theta - b(\theta) + c(y)\}$ we have $\mu = b'(\theta)$ and $v(\mu) = b''(\theta)$. The log-likelihood contribution $y_i\theta_i - b(\theta_i)$ has

$$\frac{\partial}{\partial \beta_j} = (y_i - \mu_i)\frac{\partial \theta_i}{\partial \beta_j} = (y_i - \mu_i)\frac{\partial \theta_i}{\partial \mu_i}\frac{\partial \mu_i}{\partial \beta_j} = \frac{y_i - \mu_i}{v(\mu_i)}\frac{\partial \mu_i}{\partial \beta_j},$$

since $\partial \mu / \partial \theta = b''(\theta) = v(\mu)$. Summing over i gives (11.7).

(b) $\partial/\partial\beta_j$ of $\sum_i (y_i-\mu_i)^2/v_i$ is $-2\sum_i (y_i-\mu_i)v_i^{-1}\partial\mu_i/\partial\beta_j$.

(c) Chain rule. ■

Wedderburn (1974) proposed using (11.7) for any link and variance function, whether or not they correspond to an exponential family member.

11.4.1 Properties of quasi-likelihood estimators

Definition (Unbiased estimating function)

A function $\mathbf{h}(\mathbf{y};\boldsymbol{\beta})$ is an **unbiased estimating function** if $E_{\boldsymbol{\beta}}[\mathbf{h}(\mathbf{Y};\boldsymbol{\beta})]=\mathbf{0}$ for all $\boldsymbol{\beta}$; the equations $\mathbf{h}(\mathbf{y};\boldsymbol{\beta})=\mathbf{0}$ that define $\hat{\boldsymbol{\beta}}$ are **estimating equations**.

The quasi-score (11.7) is unbiased as soon as $E(y_i)=\mu_i(\boldsymbol{\beta})$, whatever the true variance. It is treated as the derivative of a *quasi-log-likelihood*, which need not be a proper log-likelihood; nevertheless QL estimators behave much like ML estimators (McCullagh 1983). We now prove this rigorously, in a form that also covers GEE.

Lemma 11.8 (Asymptotics of estimating equations)

Let clusters $i=1,\dots,n$ be independent, $\boldsymbol{\beta}\in\Theta\subseteq\mathbb{R}^p$ open, with true value $\boldsymbol{\beta}_0$, and let $\boldsymbol{\gamma}$ be a nuisance parameter estimated by $\hat{\boldsymbol{\gamma}}\xrightarrow{p}\boldsymbol{\gamma}^*$. Let $\mathbf{U}_n(\boldsymbol{\beta};\boldsymbol{\gamma})=\sum_{i=1}^n \boldsymbol{\psi}_i(\boldsymbol{\beta};\boldsymbol{\gamma})$ and $\mathbf{H}_i=-\partial\boldsymbol{\psi}_i/\partial\boldsymbol{\beta}'$. Assume:
(E1) $E\,\boldsymbol{\psi}_i(\boldsymbol{\beta}_0;\boldsymbol{\gamma}^*)=\mathbf{0}$ for all i;
(E2) $n^{-1}\sum_i \operatorname{Cov}\big(\boldsymbol{\psi}_i(\boldsymbol{\beta}_0;\boldsymbol{\gamma}^*)\big)\to\mathbf{B}$, positive definite, and $\sup_i E\|\boldsymbol{\psi}_i(\boldsymbol{\beta}_0;\boldsymbol{\gamma}^*)\|^{2+\delta}<\infty$ for some $\delta>0$;
(E3) $\boldsymbol{\psi}_i$ is continuously differentiable in $\boldsymbol{\beta}$, and on neighbourhoods $\mathcal{N}$ of $\boldsymbol{\beta}_0$ and $\mathcal{M}$ of $\boldsymbol{\gamma}^*$, $\sup_{\mathcal{N}\times\mathcal{M}}\|n^{-1}\sum_i \mathbf{H}_i(\boldsymbol{\beta};\boldsymbol{\gamma})-\mathbf{A}(\boldsymbol{\beta},\boldsymbol{\gamma})\|\xrightarrow{p}0$ for a continuous $\mathbf{A}(\cdot,\cdot)$ with $\mathbf{A}=\mathbf{A}(\boldsymbol{\beta}_0,\boldsymbol{\gamma}^*)$ non-singular;
(E4) $n^{-1/2}[\mathbf{U}_n(\boldsymbol{\beta}_0;\hat{\boldsymbol{\gamma}})-\mathbf{U}_n(\boldsymbol{\beta}_0;\boldsymbol{\gamma}^*)]\xrightarrow{p}\mathbf{0}$.
Then:
(i) for every sufficiently small $r>0$, with probability tending to one, $\mathbf{U}_n(\boldsymbol{\beta};\hat{\boldsymbol{\gamma}})=\mathbf{0}$ has exactly one root $\hat{\boldsymbol{\beta}}$ in the ball $\|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|\le r$, and $\hat{\boldsymbol{\beta}}\xrightarrow{p}\boldsymbol{\beta}_0$;
(ii) $\sqrt{n}(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}_0)\xrightarrow{d}N(\mathbf{0},\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1\prime})$;
(iii) if moreover $n^{-1}\sum_i \boldsymbol{\psi}_i\boldsymbol{\psi}_i'$ converges in probability, uniformly on $\mathcal{N}\times\mathcal{M}$, to a continuous limit equal to $\mathbf{B}$ at $(\boldsymbol{\beta}_0,\boldsymbol{\gamma}^*)$, then with $\hat{\mathbf{A}}=n^{-1}\sum_i \mathbf{H}_i(\hat{\boldsymbol{\beta}};\hat{\boldsymbol{\gamma}})$ and $\hat{\mathbf{B}}=n^{-1}\sum_i \boldsymbol{\psi}_i\boldsymbol{\psi}_i'(\hat{\boldsymbol{\beta}};\hat{\boldsymbol{\gamma}})$, the sandwich $\hat{\mathbf{A}}^{-1}\hat{\mathbf{B}}\hat{\mathbf{A}}^{-1\prime}\xrightarrow{p}\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1\prime}$.

Proof. [Supplement]

Step 1. Set-up. Let $c=\|\mathbf{A}^{-1}\|$. By continuity of $\mathbf{A}(\cdot,\cdot)$ choose $r>0$ and a neighbourhood $\mathcal{M}'\subseteq\mathcal{M}$ of $\boldsymbol{\gamma}^*$ with $\bar{B}_r=\{\|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|\le r\}\subseteq\mathcal{N}$ and $\|\mathbf{A}(\boldsymbol{\beta},\boldsymbol{\gamma})-\mathbf{A}\|\le 1/(4c)$ on $\bar{B}_r\times\mathcal{M}'$. Let ε_n be the supremum in (E3). Consider the event

$$E_n=\Big\{\hat{\boldsymbol{\gamma}}\in\mathcal{M}',\ \ \varepsilon_n\le\tfrac{1}{4c},\ \ \|n^{-1}\mathbf{U}_n(\boldsymbol{\beta}_0;\hat{\boldsymbol{\gamma}})\|\le\tfrac{r}{2c}\Big\}.$$

By (E1)–(E2) and Chebyshev's inequality, $n^{-1}\mathbf{U}_n(\boldsymbol{\beta}_0;\boldsymbol{\gamma}^*)\xrightarrow{p}\mathbf{0}$; with (E4) also $n^{-1}\mathbf{U}_n(\boldsymbol{\beta}_0;\hat{\boldsymbol{\gamma}})\xrightarrow{p}\mathbf{0}$. With $\hat{\boldsymbol{\gamma}}\xrightarrow{p}\boldsymbol{\gamma}^*$ and $\varepsilon_n\xrightarrow{p}0$, $P(E_n)\to1$.

Step 2. Linearization. For $\boldsymbol{\beta}\in\bar{B}_r$ (a convex set), the fundamental theorem of calculus along the segment from $\boldsymbol{\beta}_0$ to $\boldsymbol{\beta}$ gives

$$n^{-1}\mathbf{U}_n(\boldsymbol{\beta};\hat{\boldsymbol{\gamma}})=n^{-1}\mathbf{U}_n(\boldsymbol{\beta}_0;\hat{\boldsymbol{\gamma}})-\mathbf{A}_n(\boldsymbol{\beta})(\boldsymbol{\beta}-\boldsymbol{\beta}_0),\qquad \mathbf{A}_n(\boldsymbol{\beta})=\int_0^1 n^{-1}\sum_i \mathbf{H}_i(\boldsymbol{\beta}_0+s(\boldsymbol{\beta}-\boldsymbol{\beta}_0);\hat{\boldsymbol{\gamma}})\,ds.$$

On E_n, $\|\mathbf{A}_n(\boldsymbol{\beta}) - \mathbf{A}\| \le \varepsilon_n + \frac{1}{4c} \le \frac{1}{2c}$ for all $\boldsymbol{\beta} \in \bar{B}_r$.

Step 3. Existence. Define $F(\boldsymbol{\beta}) = \boldsymbol{\beta} + \mathbf{A}^{-1} n^{-1} \mathbf{U}_n(\boldsymbol{\beta}; \hat{\boldsymbol{\gamma}})$; its fixed points are roots. By Step 2,

$$F(\boldsymbol{\beta}) - \boldsymbol{\beta}_0 = \mathbf{A}^{-1} n^{-1} \mathbf{U}_n(\boldsymbol{\beta}_0; \hat{\boldsymbol{\gamma}}) + \mathbf{A}^{-1}(\mathbf{A} - \mathbf{A}_n(\boldsymbol{\beta}))(\boldsymbol{\beta} - \boldsymbol{\beta}_0),$$

so on E_n, $\|F(\boldsymbol{\beta}) - \boldsymbol{\beta}_0\| \le c \cdot \frac{r}{2c} + c \cdot \frac{1}{2c} \cdot r = r$. Thus the continuous map F sends $\bar{B}_r$ into itself, and by Brouwer's fixed-point theorem it has a fixed point $\hat{\boldsymbol{\beta}} \in \bar{B}_r$.

Step 4. Uniqueness. If $\boldsymbol{\beta}_1, \boldsymbol{\beta}_2 \in \bar{B}_r$ are roots, the same integral argument gives $\mathbf{0} = \tilde{\mathbf{A}}(\boldsymbol{\beta}_1 - \boldsymbol{\beta}_2)$ with $\|\tilde{\mathbf{A}} - \mathbf{A}\| \le \frac{1}{2c}$. Then $\|\mathbf{A}^{-1}(\tilde{\mathbf{A}} - \mathbf{A})\| \le \frac{1}{2} < 1$, so $\tilde{\mathbf{A}} = \mathbf{A}(\mathbf{I} + \mathbf{A}^{-1}(\tilde{\mathbf{A}} - \mathbf{A}))$ is invertible (Neumann series) and $\boldsymbol{\beta}_1 = \boldsymbol{\beta}_2$.

Step 5. Consistency. The argument works for every smaller radius $r' \le r$, and by uniqueness the root in $\bar{B}_{r'}$ is the root in $\bar{B}_r$. Hence $P(\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\| \le r') \to 1$ for every $r' > 0$: $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$. This proves (i).

Step 6. Asymptotic normality. On E_n, Step 2 at $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}$ gives

$$\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) = \mathbf{A}_n(\hat{\boldsymbol{\beta}})^{-1}\, n^{-1/2} \mathbf{U}_n(\boldsymbol{\beta}_0; \hat{\boldsymbol{\gamma}}).$$

First, $\|\mathbf{A}_n(\hat{\boldsymbol{\beta}}) - \mathbf{A}\| \le \varepsilon_n + \sup_{s \in [0,1]} \|\mathbf{A}(\boldsymbol{\beta}_0 + s(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0), \hat{\boldsymbol{\gamma}}) - \mathbf{A}\| \xrightarrow{p} 0$ by (E3), continuity and consistency. Second, by (E4), $n^{-1/2}\mathbf{U}_n(\boldsymbol{\beta}_0; \hat{\boldsymbol{\gamma}}) = n^{-1/2}\mathbf{U}_n(\boldsymbol{\beta}_0; \boldsymbol{\gamma}^*) + o_p(1)$, a normalized sum of independent mean-zero vectors. For any fixed $\boldsymbol{\lambda} \neq \mathbf{0}$, Lyapunov's condition holds,

$$\frac{\sum_i E|\boldsymbol{\lambda}'\boldsymbol{\psi}_i|^{2+\delta}}{\left(\sum_i \operatorname{Var}(\boldsymbol{\lambda}'\boldsymbol{\psi}_i)\right)^{1+\delta/2}} = \frac{O(n)}{\left(n\, \boldsymbol{\lambda}'\mathbf{B}\boldsymbol{\lambda}\,(1 + o(1))\right)^{1+\delta/2}} \to 0,$$

so $n^{-1/2}\boldsymbol{\lambda}'\mathbf{U}_n \xrightarrow{d} N(0, \boldsymbol{\lambda}'\mathbf{B}\boldsymbol{\lambda})$ and, by Cramér–Wold, $n^{-1/2}\mathbf{U}_n(\boldsymbol{\beta}_0; \boldsymbol{\gamma}^*) \xrightarrow{d} N(\mathbf{0}, \mathbf{B})$. Slutsky's theorem gives (ii).

Step 7. Sandwich. $\hat{\mathbf{A}} \xrightarrow{p} \mathbf{A}$ as in Step 6 (uniform convergence plus continuity at the consistent $(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\gamma}})$), and $\hat{\mathbf{B}} \xrightarrow{p} \mathbf{B}$ by the extra assumption in (iii) in the same way. Matrix inversion and multiplication are continuous, which proves (iii). ∎

Theorem 11.9 (Consistency and asymptotic normality of QL estimators)

Suppose the mean model $g(\mu_i) = \mathbf{x}_i'\boldsymbol{\beta}$ is correct at $\boldsymbol{\beta}_0$ (the variance function v may be wrong) and that conditions (E2), (E3) and the condition in (iii) of Lemma 11.8 hold for $\boldsymbol{\psi}_i = (\partial\mu_i/\partial\boldsymbol{\beta})(y_i - \mu_i)/v(\mu_i)$ (they hold, e.g., for bounded covariates, smooth g and v, $v > 0$, and bounded fourth moments of y_i). Then the QL estimator is consistent and asymptotically normal, with

$$\operatorname{Cov}(\hat{\boldsymbol{\beta}}) \approx \mathbf{V}\Big[\sum_i \frac{\partial\mu_i}{\partial\boldsymbol{\beta}} \frac{\operatorname{Var}(Y_i)}{v(\mu_i)^2} \frac{\partial\mu_i}{\partial\boldsymbol{\beta}'}\Big]\mathbf{V}, \qquad \mathbf{V} = \Big[\sum_i \frac{\partial\mu_i}{\partial\boldsymbol{\beta}} \frac{1}{v(\mu_i)} \frac{\partial\mu_i}{\partial\boldsymbol{\beta}'}\Big]^{-1}. \tag{11.8}$$

If $\operatorname{Var}(Y_i) = v(\mu_i)$, this reduces to $\operatorname{Cov}(\hat{\boldsymbol{\beta}}) \approx \mathbf{V}$, the GLM formula.

Proof.

Step 1. (E1) holds because $E(y_i) = \mu_i(\boldsymbol{\beta}_0)$ and the coefficient $(\partial\mu_i/\partial\boldsymbol{\beta})/v(\mu_i)$ is nonrandom; the true variance plays no role. There is no nuisance parameter, so (E4) is vacuous.

Step 2. Differentiating,

$$\mathbf{H}_i(\boldsymbol{\beta}) = \frac{1}{v(\mu_i)} \frac{\partial\mu_i}{\partial\boldsymbol{\beta}} \frac{\partial\mu_i}{\partial\boldsymbol{\beta}'} - \frac{\partial}{\partial\boldsymbol{\beta}'}\Big[\frac{1}{v(\mu_i)} \frac{\partial\mu_i}{\partial\boldsymbol{\beta}}\Big](y_i - \mu_i).$$

At $\boldsymbol{\beta}_0$ the second term has mean zero, so $\mathbf{A} = \lim n^{-1} \sum_i v(\mu_i)^{-1}(\partial\mu_i/\partial\boldsymbol{\beta})(\partial\mu_i/\partial\boldsymbol{\beta}')$. Also $\operatorname{Cov}(\boldsymbol{\psi}_i) = v(\mu_i)^{-2} \operatorname{Var}(Y_i)(\partial\mu_i/\partial\boldsymbol{\beta})(\partial\mu_i/\partial\boldsymbol{\beta}')$, giving $\mathbf{B}$.

Step 3. Lemma 11.8(ii) gives $\mathrm{Cov}(\hat{\boldsymbol{\beta}}) \approx n^{-1}\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$, which is (11.8) after writing $n\mathbf{A} \approx \mathbf{V}^{-1}$ and $n\mathbf{B} \approx \sum_i(\cdots)$. If $\mathrm{Var}(Y_i) = v(\mu_i)$, then $\mathbf{B} = \mathbf{A}$ and the sandwich collapses to $\mathbf{V}$. ■

The book's heuristic is the essence of Step 1: $\mathbf{u}(\boldsymbol{\beta})/n$ is a vector of sample means whose expectation vanishes at the true $\boldsymbol{\beta}$ whatever the variance, and $\hat{\boldsymbol{\beta}}$ is the point where it vanishes.

11.4.2 GEE methodology: technical details

For cluster i let $\mathbf{y}_i = (y_{i1}, \ldots, y_{iT_i})'$, $\boldsymbol{\mu}_i = E(\mathbf{Y}_i)$, $\mathbf{x}_{it}$ the $p \times 1$ predictor vector for y_{it} (it may vary over t), $\mathbf{X}_i$ the $T_i \times p$ matrix with rows $\mathbf{x}_{it}'$, and $\eta_{it} = g(\mu_{it}) = \mathbf{x}_{it}'\boldsymbol{\beta}$. The model refers to each margin. Suppose

$$f(y_{it}; \theta_{it}, \phi) = \exp\{[y_{it}\theta_{it} - b(\theta_{it})]/\phi + c(y_{it}, \phi)\},$$

so $\mu_{it} = b'(\theta_{it})$ and $\mathrm{Var}(Y_{it}) = \phi\, b''(\theta_{it})$.

Definition (Generalized estimating equations)

Let $\mathbf{B}_i = \mathrm{diag}\,(b''(\theta_{i1}), \ldots, b''(\theta_{iT_i}))$ and let $\mathbf{R}(\boldsymbol{\alpha})$ be a working correlation matrix. The **working covariance** is

$$\mathbf{V}_i = \phi\, \mathbf{B}_i^{1/2}\mathbf{R}(\boldsymbol{\alpha})\mathbf{B}_i^{1/2} \tag{11.9}$$

($= \mathrm{Cov}(\mathbf{Y}_i)$ if $\mathbf{R}$ is the true correlation). Let $\boldsymbol{\Delta}_i = \mathrm{diag}(\partial\theta_{it}/\partial\eta_{it})$ (the identity for the canonical link) and $\mathbf{D}_i = \partial\boldsymbol{\mu}_i/\partial\boldsymbol{\beta}' = \mathbf{B}_i\boldsymbol{\Delta}_i\mathbf{X}_i$ ($T_i \times p$). The GEE estimator $\hat{\boldsymbol{\beta}}$ solves

$$\mathbf{u}(\boldsymbol{\beta}) = \sum_{i=1}^{n} \mathbf{D}_i'\mathbf{V}_i^{-1}(\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}. \tag{11.10}$$

The formula $\mathbf{D}_i = \mathbf{B}_i\boldsymbol{\Delta}_i\mathbf{X}_i$ is the chain rule $\partial\mu_{it}/\partial\beta_j = (\partial\mu_{it}/\partial\theta_{it})(\partial\theta_{it}/\partial\eta_{it})(\partial\eta_{it}/\partial\beta_j)$. Since ϕ is a common scalar factor of $\mathbf{V}_i$, it cancels from (11.10): $\hat{\boldsymbol{\beta}}$ does not depend on ϕ (given $\boldsymbol{\alpha}$).

Proposition 11.10 (Independence working correlation gives the GLM fit)

With $\mathbf{R} = \mathbf{I}$, (11.10) becomes $\sum_i \mathbf{X}_i'\boldsymbol{\Delta}_i(\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$ (for the canonical link, $\sum_i \mathbf{X}_i'(\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$): the ordinary GLM likelihood equations treating all y_{it} as independent.

Proof. With $\mathbf{R} = \mathbf{I}$, $\mathbf{V}_i = \phi\mathbf{B}_i$, so $\mathbf{D}_i'\mathbf{V}_i^{-1} = \mathbf{X}_i'\boldsymbol{\Delta}_i\mathbf{B}_i(\phi\mathbf{B}_i)^{-1} = \phi^{-1}\mathbf{X}_i'\boldsymbol{\Delta}_i$ (diagonal matrices commute). Dropping the factor $1/\phi$ gives the stated equations, which are the GLM likelihood equations of Proposition 11.7(a) summed over all (i, t). ■

Definition (Moment estimators of the working parameters)

With Pearson residuals $e_{it} = (y_{it} - \hat{\mu}_{it})/b''(\hat{\theta}_{it})^{1/2}$, $N = \sum_i T_i$ (Liang and Zeger 1986):

$$\hat{\phi} = \frac{1}{N - p}\sum_i\sum_t e_{it}^2, \qquad \hat{\alpha}_{\mathrm{exch}} = \frac{\sum_i \sum_{t<s} e_{it}e_{is}}{\hat{\phi}\,[\sum_i T_i(T_i - 1)/2 - p]},$$

$$\hat{\alpha}_{\mathrm{AR1}} = \frac{\sum_i \sum_{t \le T_i - 1} e_{it}e_{i,t+1}}{\hat{\phi}\,[\sum_i (T_i - 1) - p]}, \qquad \hat{R}_{ts} = \frac{1}{n\hat{\phi}}\sum_i e_{it}e_{is} \quad \text{(unstructured)}.$$

The fitting algorithm alternates a modified Fisher scoring step for $\boldsymbol{\beta}$ given $(\boldsymbol{\alpha}, \phi)$,

$$\boldsymbol{\beta}^{(k+1)} = \boldsymbol{\beta}^{(k)} + \Big[\sum_i \mathbf{D}_i'\mathbf{V}_i^{-1}\mathbf{D}_i\Big]^{-1}\sum_i \mathbf{D}_i'\mathbf{V}_i^{-1}(\mathbf{y}_i - \boldsymbol{\mu}_i),$$

with moment updates of $(\boldsymbol{\alpha}, \phi)$ from the current residuals. Other algorithms solve estimating equations for $\boldsymbol{\beta}$ and $\boldsymbol{\alpha}$ jointly. GEE algorithms need not converge, but one iteration is often adequate. For time series, AR(1) treats observations farther apart in time as more weakly correlated.

Theorem 11.11 (Liang–Zeger: consistency and asymptotic normality of GEE)

Assume:
(G1) the marginal mean model is correct: $E(\mathbf{Y}_i) = \boldsymbol{\mu}_i(\boldsymbol{\beta}_0)$ for all i (the working correlation may be wrong);
(G2) $\sqrt{n}(\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}^*) = O_p(1)$ for some $\boldsymbol{\alpha}^*$ (not necessarily a "true" value);
(G3) conditions (E2), (E3) and the condition in (iii) of Lemma 11.8 hold for $\boldsymbol{\psi}_i(\boldsymbol{\beta}; \boldsymbol{\alpha}) = \mathbf{D}_i' \mathbf{V}_i^{-1}(\boldsymbol{\alpha})(\mathbf{y}_i - \boldsymbol{\mu}_i)$, and $n^{-1} \sum_i \partial \boldsymbol{\psi}_i(\boldsymbol{\beta}_0; \boldsymbol{\alpha})/\partial \boldsymbol{\alpha}' \xrightarrow{p} \mathbf{0}$ uniformly for $\boldsymbol{\alpha}$ near $\boldsymbol{\alpha}^*$.
Then, as the number of clusters $n \to \infty$, $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$ and $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{V}_G)$, where $\mathbf{V}_G = \lim_n \mathbf{V}_{G,n}$,

$$\mathbf{V}_{G,n} = n \Big[\sum_i \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i \Big]^{-1} \Big[\sum_i \mathbf{D}_i' \mathbf{V}_i^{-1} \operatorname{Cov}(\mathbf{Y}_i) \mathbf{V}_i^{-1} \mathbf{D}_i \Big] \Big[\sum_i \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i \Big]^{-1},$$

evaluated at $(\boldsymbol{\beta}_0, \boldsymbol{\alpha}^*)$. The sandwich estimator $\hat{\mathbf{V}}_{G,n}/n$, which substitutes $\hat{\boldsymbol{\beta}}$, $\hat{\boldsymbol{\alpha}}$, $\hat{\phi}$ and replaces $\operatorname{Cov}(\mathbf{Y}_i)$ by $(\mathbf{y}_i - \hat{\boldsymbol{\mu}}_i)(\mathbf{y}_i - \hat{\boldsymbol{\mu}}_i)'$, is consistent for $\operatorname{Cov}(\hat{\boldsymbol{\beta}})$.

Proof. [Supplement] We verify the hypotheses of Lemma 11.8 with nuisance $\boldsymbol{\gamma} = \boldsymbol{\alpha}$ (ϕ cancels from (11.10)).

Step 1. (E1). At $\boldsymbol{\beta}_0$, $\mathbf{D}_i$ and $\mathbf{V}_i(\boldsymbol{\alpha}^*)$ are nonrandom and $E(\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$ by (G1). Hence $E\boldsymbol{\psi}_i(\boldsymbol{\beta}_0; \boldsymbol{\alpha}^*) = \mathbf{0}$ *for any* $\boldsymbol{\alpha}^*$; this is why a wrong working correlation does not destroy consistency.

Step 2. The matrices $\mathbf{A}$ and $\mathbf{B}$. Differentiating,

$$\mathbf{H}_i = \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i - \sum_j \Big[\frac{\partial (\mathbf{D}_i' \mathbf{V}_i^{-1})}{\partial \beta_j} (\mathbf{y}_i - \boldsymbol{\mu}_i) \Big] \mathbf{e}_j',$$

and the second term has mean zero at $\boldsymbol{\beta}_0$, so $\mathbf{A} = \lim n^{-1} \sum_i \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i$. Also $\operatorname{Cov}(\boldsymbol{\psi}_i) = \mathbf{D}_i' \mathbf{V}_i^{-1} \operatorname{Cov}(\mathbf{Y}_i) \mathbf{V}_i^{-1} \mathbf{D}_i$, whose average tends to $\mathbf{B}$. (E2), (E3) hold by (G3).

Step 3. (E4). By the mean value theorem applied to each component, for some $\bar{\boldsymbol{\alpha}}$ between $\hat{\boldsymbol{\alpha}}$ and $\boldsymbol{\alpha}^*$,

$$n^{-1/2} [\mathbf{U}_n(\boldsymbol{\beta}_0; \hat{\boldsymbol{\alpha}}) - \mathbf{U}_n(\boldsymbol{\beta}_0; \boldsymbol{\alpha}^*)] = \Big[n^{-1} \sum_i \frac{\partial \boldsymbol{\psi}_i(\boldsymbol{\beta}_0; \bar{\boldsymbol{\alpha}})}{\partial \boldsymbol{\alpha}'} \Big] \sqrt{n}(\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}^*).$$

Here $\partial \boldsymbol{\psi}_i / \partial \alpha_k = \mathbf{D}_i' (\partial \mathbf{V}_i^{-1} / \partial \alpha_k)(\mathbf{y}_i - \boldsymbol{\mu}_i)$ has mean zero at $\boldsymbol{\beta}_0$, and by (G3) the bracket is $o_p(1)$, while $\sqrt{n}(\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}^*) = O_p(1)$ by (G2). The product is $o_p(1)$. Estimating $\boldsymbol{\alpha}$ thus has no first-order effect on $\hat{\boldsymbol{\beta}}$.

Step 4. Lemma 11.8(i)–(ii) gives consistency and $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1})$, and $\mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1} = \lim \mathbf{V}_{G,n}$. Lemma 11.8(iii) gives consistency of the sandwich, since $\hat{\mathbf{B}} = n^{-1} \sum_i \mathbf{D}_i' \mathbf{V}_i^{-1} (\mathbf{y}_i - \hat{\boldsymbol{\mu}}_i)(\mathbf{y}_i - \hat{\boldsymbol{\mu}}_i)' \mathbf{V}_i^{-1} \mathbf{D}_i$ evaluated at $(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\alpha}})$; replacing $\hat{\mathbf{A}}$ by the expected-information version $n^{-1} \sum \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i$ at the estimates changes it by a term that is $o_p(1)$ (Step 2). ∎

Corollary 11.12 (Model-based covariance)

If the working covariance is correct, $\operatorname{Cov}(\mathbf{Y}_i) = \mathbf{V}_i$, then $\mathbf{V}_{G,n}/n = \big(\sum_i \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i \big)^{-1}$.

Proof. Substitute $\operatorname{Cov}(\mathbf{Y}_i) = \mathbf{V}_i$ in the middle factor: it becomes $\sum_i \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i$ and cancels one outer inverse. ∎

Theorem 11.13 (Optimality of the true covariance as weight)

Consider all estimating equations $\sum_i \mathbf{D}_i'\mathbf{W}_i(\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$ with nonrandom positive definite weights $\mathbf{W}_i$, and let $\boldsymbol{\Sigma}_i = \mathrm{Cov}(\mathbf{Y}_i)$. Their asymptotic covariance satisfies, in the Loewner order,

$$\mathbf{G}^{-1}\mathbf{M}\mathbf{G}^{-1} \;\geq\; \mathbf{F}^{-1}, \qquad \mathbf{G} = \sum_i \mathbf{D}_i'\mathbf{W}_i\mathbf{D}_i, \quad \mathbf{M} = \sum_i \mathbf{D}_i'\mathbf{W}_i\boldsymbol{\Sigma}_i\mathbf{W}_i\mathbf{D}_i, \quad \mathbf{F} = \sum_i \mathbf{D}_i'\boldsymbol{\Sigma}_i^{-1}\mathbf{D}_i,$$

with equality when $\mathbf{W}_i = \boldsymbol{\Sigma}_i^{-1}$. So a correct working covariance gives the most efficient GEE estimator.

Proof. [Supplement]

Step 1. For each i, the matrix

$$\begin{pmatrix} \mathbf{D}_i'\mathbf{W}_i \\ \mathbf{D}_i'\boldsymbol{\Sigma}_i^{-1} \end{pmatrix} \boldsymbol{\Sigma}_i \begin{pmatrix} \mathbf{W}_i\mathbf{D}_i & \boldsymbol{\Sigma}_i^{-1}\mathbf{D}_i \end{pmatrix} = \begin{pmatrix} \mathbf{D}_i'\mathbf{W}_i\boldsymbol{\Sigma}_i\mathbf{W}_i\mathbf{D}_i & \mathbf{D}_i'\mathbf{W}_i\mathbf{D}_i \\ \mathbf{D}_i'\mathbf{W}_i\mathbf{D}_i & \mathbf{D}_i'\boldsymbol{\Sigma}_i^{-1}\mathbf{D}_i \end{pmatrix}$$

is positive semidefinite (it has the form $\mathbf{K}'\boldsymbol{\Sigma}_i\mathbf{K}$). Summing over i, $\left(\begin{smallmatrix} \mathbf{M} & \mathbf{G} \\ \mathbf{G} & \mathbf{F} \end{smallmatrix}\right) \geq 0$.

Step 2. Since $\mathbf{F} > 0$, the Schur complement is positive semidefinite: $\mathbf{M} - \mathbf{G}\mathbf{F}^{-1}\mathbf{G} \geq 0$.

Step 3. Multiplying on both sides by the symmetric $\mathbf{G}^{-1}$ preserves the order: $\mathbf{G}^{-1}\mathbf{M}\mathbf{G}^{-1} \geq \mathbf{F}^{-1}$. If $\mathbf{W}_i = \boldsymbol{\Sigma}_i^{-1}$ then $\mathbf{G} = \mathbf{M} = \mathbf{F}$ and equality holds. ■

Correlation is an awkward association measure for binary data

For binary Y_t, Y_s with $p_t = P(Y_t = 1)$, $p_s = P(Y_s = 1)$, the Fréchet bounds $\max(0, p_t + p_s - 1) \leq P(Y_t = Y_s = 1) \leq \min(p_t, p_s)$ (the first because $P(A \cap B) \geq P(A) + P(B) - 1$, the second because $A \cap B \subseteq A, B$) restrict

$$\mathrm{Corr}(Y_t, Y_s) = \frac{P(Y_t = Y_s = 1) - p_t p_s}{\sqrt{p_t(1-p_t)p_s(1-p_s)}}$$

to a range depending on the means. For instance, with $p_t = 0.1$, $p_s = 0.5$ the correlation cannot exceed $(0.1 - 0.05)/\sqrt{0.09 \cdot 0.25} = 1/3$. An alternative models within-cluster association by odds ratios (e.g. exchangeable log odds ratios), which are variation-independent of the means; the *alternating logistic regressions* algorithm (Carey, Zeger and Diggle 1993) alternates a GEE step for $\boldsymbol{\beta}$ with a logistic regression step for the log odds ratios, useful when the association is itself of interest.

11.5 Markov chains: transitional modeling

Motivation. Instead of modelling margins, one can model each response given the earlier ones. The joint distribution then factors into transition probabilities, and fitting reduces to ordinary multinomial or logistic models for the transitions.

For responses Y_t at times $t = 0, 1, 2, \ldots$, the family $\{Y_t\}$ is a stochastic process; its *state space* is the set of possible values, and Y_0 is the initial state. Here state space and time are discrete.

11.5.1 Transitional models

Transitional models describe Y_t in terms of the previous responses $y_0, \ldots, y_{t-1}$ (and covariates), via the factorization

$$f(y_0, \ldots, y_T) = f(y_0) f(y_1 \mid y_0) f(y_2 \mid y_0, y_1) \cdots f(y_T \mid y_0, \ldots, y_{T-1}).$$

Unlike marginal models, they condition on past responses.

11.5.2 First-order Markov chains

Definition (Markov chain)

$\{Y_t\}$ is a (first-order) **Markov chain** if, for all t, Y_{t+1} given $(Y_0, \dots, Y_t)$ has the same distribution as Y_{t+1} given Y_t alone. It is a **kth-order** chain if Y_{t+1} given $(Y_0, \dots, Y_t)$ depends only on $(Y_{t-k+1}, \dots, Y_t)$. The **transition probabilities** are $\pi_{j|i}(t) = P(Y_t = j \mid Y_{t-1} = i)$, with $\sum_j \pi_{j|i}(t) = 1$; they form the one-step $I \times I$ transition matrix.

Theorem 11.14 (Markov factorization and its loglinear form)

(a) $\{Y_0, \dots, Y_T\}$ is a Markov chain if and only if

$$f(y_0, \dots, y_T) = f(y_0) f(y_1 \mid y_0) f(y_2 \mid y_1) \cdots f(y_T \mid y_{T-1}). \tag{11.11}$$

(b) A positive joint distribution is that of a Markov chain if and only if it satisfies the loglinear model $(Y_0Y_1, Y_1Y_2, \dots, Y_{T-1}Y_T)$, i.e. $\log \pi(y_0, \dots, y_T) = \sum_{t=1}^{T} h_t(y_{t-1}, y_t)$ for some functions h_t.

Proof. [Supplement]

Step 1. (a) The general factorization of Section 11.5.1 holds always; the Markov property replaces each $f(y_t \mid y_0, \dots, y_{t-1})$ by $f(y_t \mid y_{t-1})$, giving (11.11). Conversely, if (11.11) holds, summing it over $y_{t+1}, \dots, y_T$ (each later factor sums to 1, from the last backwards) gives $f(y_0, \dots, y_t) = f(y_0) \prod_{s \le t} f(y_s \mid y_{s-1})$, and dividing the expressions for $t+1$ and t gives $f(y_{t+1} \mid y_0, \dots, y_t) = f(y_{t+1} \mid y_t)$.

Step 2. (b, $\Rightarrow$) Taking logs of (11.11) and absorbing $\log f(y_0)$ into h_1 gives $\sum_t h_t(y_{t-1}, y_t)$, which contains only main effects and two-factor terms for consecutive pairs.

Step 3. (b, $\Leftarrow$) Suppose $\pi(\mathbf{y}) = \prod_{t=1}^{T} g_t(y_{t-1}, y_t)$ with $g_t = e^{h_t}$. Define $G_t(y_t) = \sum_{y_{t+1}, \dots, y_T} \prod_{s > t} g_s(y_{s-1}, y_s)$ (with $G_T \equiv 1$). Then

$$P(y_{t+1} \mid y_0, \dots, y_t) = \frac{\prod_{s \le t} g_s \cdot g_{t+1}(y_t, y_{t+1}) G_{t+1}(y_{t+1})}{\prod_{s \le t} g_s \cdot G_t(y_t)} = \frac{g_{t+1}(y_t, y_{t+1}) G_{t+1}(y_{t+1})}{G_t(y_t)},$$

which depends only on (y_t, y_{t+1}): the Markov property. ∎

Thus fitting the loglinear model $(Y_0Y_1, \dots, Y_{T-1}Y_T)$ to the I^{T+1} table of sequences tests the Markov property; adding three-factor terms $(Y_{t-2}Y_{t-1}Y_t)$ gives a second-order chain.

Theorem 11.15 (ML estimation of transition probabilities)

Let $n_{ij}(t)$ be the number of transitions from state i at time $t-1$ to state j at time t, $n_{i+}(t) = \sum_j n_{ij}(t)$, and $\{n_{i0}\}$ the initial counts, multinomial with probabilities $\{\pi_{i0}\}$. For n independent subjects following a Markov chain, the likelihood is proportional to

$$\Big[\prod_{i=1}^{I} \pi_{i0}^{n_{i0}}\Big] \prod_{t=1}^{T} \prod_{i=1}^{I} \prod_{j=1}^{I} \pi_{j|i}(t)^{n_{ij}(t)}, \tag{11.12}$$

and the ML estimates are $\hat{\pi}_{j|i}(t) = n_{ij}(t)/n_{i+}(t)$ (Anderson and Goodman 1957). For a time-homogeneous chain, $\pi_{j|i}(t) \equiv \pi_{j|i}$, the ML estimate is $\hat{\pi}_{j|i} = \sum_t n_{ij}(t) / \sum_t n_{i+}(t)$.

Proof.

Step 1. By (11.11) each subject's probability is $\pi_{y_0 0}\prod_t \pi_{y_t|y_{t-1}}(t)$; multiplying over subjects and grouping identical factors gives (11.12).

Step 2. [Supplement] The likelihood is a product of separate kernels, one for $\{\pi_{i0}\}$ and one for each (t,i), the latter being the multinomial kernel $\prod_j \pi_{j|i}(t)^{n_{ij}(t)}$ with $\sum_j \pi_{j|i}(t)=1$; the parameters of different kernels are unconstrained by each other, so each is maximized separately.

Step 3. For one kernel $\sum_j n_j \log \pi_j$ with $\sum_j \pi_j = 1$ and $m=\sum_j n_j > 0$, Gibbs' inequality gives

$$\sum_j n_j \log \pi_j - \sum_j n_j \log \frac{n_j}{m} = m \sum_j \frac{n_j}{m} \log \frac{\pi_j}{n_j/m} \le m \log \sum_j \pi_j = 0$$

(Jensen, concavity of log), with equality iff $\pi_j = n_j/m$. So $\hat\pi_{j|i}(t) = n_{ij}(t)/n_{i+}(t)$.

Step 4. Time-homogeneous case: the kernels for fixed i combine into $\prod_j \pi_{j|i}^{\sum_t n_{ij}(t)}$, and Step 3 applies with $n_j = \sum_t n_{ij}(t)$. ■

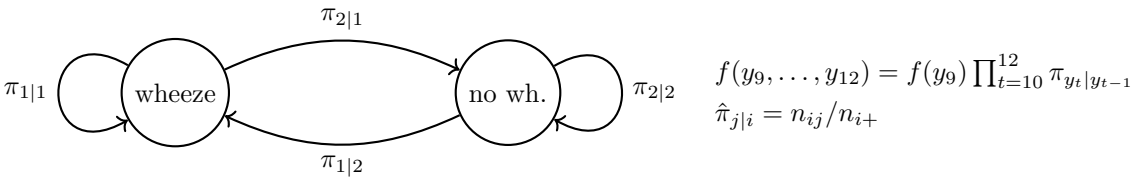


Figure 11.2. A two-state first-order Markov chain; each transition is a multinomial (here binomial) trial.

11.5.3 Respiratory illness example

Example 11.16 (Wheeze at ages 9–12)

In a Harvard study of air pollution, children were examined annually at ages 9–12 for wheeze (1 = wheeze, 2 = no wheeze):

Y_9	Y_{10}	Y_{11}	Y_{12}	Count	Y_9	Y_{10}	Y_{11}	Y_{12}	Count
1	1	1	1	94	2	1	1	1	19
1	1	1	2	30	2	1	1	2	15
1	1	2	1	15	2	1	2	1	10
1	1	2	2	28	2	1	2	2	44
1	2	1	1	14	2	2	1	1	17
1	2	1	2	9	2	2	1	2	42
1	2	2	1	12	2	2	2	1	35
1	2	2	2	63	2	2	2	2	572

The first-order chain $(Y_9Y_{10}, Y_{10}Y_{11}, Y_{11}Y_{12})$ fits poorly ($G^2 = 122.9$, df $= 8$), and so does the second-order chain $(Y_9Y_{10}Y_{11}, Y_{10}Y_{11}Y_{12})$ ($G^2 = 23.9$, df $= 4$); the poor fits may partly reflect subject heterogeneity (ignored covariates such as parental smoking). The model with all six pairwise associations fits well ($G^2 = 1.5$, df $= 5$). Its conditional log odds ratios are 1.81, 1.65, 1.85 for the pairs one year apart $(9,10),(10,11),(11,12)$ and 0.95, 1.05, 1.07 for $(9,11),(9,12),(10,12)$. The simpler model with one common value for each group fits well ($G^2 = 2.3$, df $= 9$), with estimates 1.75 (one year apart) and 1.04 (more than one year apart).

11.5.4 Transitional models with explanatory variables

With covariates $\mathbf{x}$,

$$f(y_1, \ldots, y_T; \mathbf{x}) = f(y_1; \mathbf{x}) f(y_2 \mid y_1; \mathbf{x}) \cdots f(y_T \mid y_1, \ldots, y_{T-1}; \mathbf{x}),$$

and for binary y one may specify a logistic model for each factor:

$$f(y_t \mid y_1, \ldots, y_{t-1}; \mathbf{x}_t) = \frac{\exp\left[y_t(\alpha + \beta_1 y_1 + \cdots + \beta_{t-1} y_{t-1} + \boldsymbol{\beta}' \mathbf{x}_t)\right]}{1 + \exp(\alpha + \beta_1 y_1 + \cdots + \beta_{t-1} y_{t-1} + \boldsymbol{\beta}' \mathbf{x}_t)}, \qquad y_t = 0, 1,$$

a **regressive logistic model** (Bonney 1987): previous responses act as explanatory variables, and $\mathbf{x}_t$ may change with t. First-order Markov structure sets the coefficients of $y_1, \ldots, y_{t-2}$ to zero. Interaction between $\mathbf{x}_t$ and y_{t-1} may be needed. Note that the meaning and size of $\boldsymbol{\beta}$ depend on how many previous responses are in the model: within-cluster effects can shrink markedly once we condition on the past, unlike marginal model effects, whose interpretation does not depend on the dependence structure.

Proposition 11.17 (Fitting transitional models by ordinary logistic regression)

Conditional on the first response(s), the likelihood of a regressive logistic model is the product, over subjects and transitions, of Bernoulli likelihoods with the stated logits. Hence the conditional ML estimates are obtained by fitting an ordinary logistic regression that treats each transition as a separate independent observation, with previous responses as predictors.

Proof. [Supplement] By the factorization, subject i contributes

$$f(y_{i1}; \mathbf{x}) \prod_{t \geq 2} f(y_{it} \mid y_{i1}, \ldots, y_{i,t-1}; \mathbf{x}_{it}).$$

Ignoring (conditioning on) the first factor, each remaining factor is a Bernoulli probability $\text{expit}(\eta_{it})^{y_{it}}[1 - \text{expit}(\eta_{it})]^{1-y_{it}}$, where η_{it} is linear in the parameters with known regressors (the observed past responses and $\mathbf{x}_{it}$). Independence across subjects makes the conditional likelihood a product of such terms over all (i, t), which is exactly the likelihood of a logistic regression with one observation per transition. ■

Although transitions of one subject are dependent, the conditional likelihood has this independent-data form, so GLM software gives correct ML estimates and (likelihood-based) standard errors.

11.5.5 Child's respiratory illness and maternal smoking

Example 11.18 (Respiratory illness and maternal smoking)

Children were evaluated annually at ages 7–10 for respiratory illness; $s = 1$ if the mother smoked regularly at the start of the study:

			No maternal smoking		Maternal smoking	
Age 7	Age 8	Age 9	Age 10 No	Age 10 Yes	Age 10 No	Age 10 Yes
No	No	No	237	10	118	6
No	No	Yes	15	4	8	2
No	Yes	No	16	2	11	1
No	Yes	Yes	7	3	6	4
Yes	No	No	24	3	7	3
Yes	No	Yes	3	2	3	1
Yes	Yes	No	6	2	4	2
Yes	Yes	Yes	5	11	4	7

The regressive logistic model

$$\text{logit}\, P(Y_t = 1) = \alpha + \beta_1 s + \beta_2 t + \beta_3 y_{t-1}, \qquad t = 8, 9, 10,$$

uses three transitions per child; the data form 12 binomials for the $2 \times 3 \times 2$ combinations of (s, t, y_{t-1}) (e.g. for $(0, 8, 0)$: $y_8 = 0$ for $237 + 10 + 15 + 4 = 266$ children and $y_8 = 1$ for $16 + 2 + 7 + 3 = 28$). The ML fit is

$$\text{logit}\, \hat{P}(Y_t = 1) = -0.293 + 0.296 s - 0.243 t + 2.211 y_{t-1},$$

with SEs $(0.846, 0.156, 0.095, 0.158)$. The previous response has a strong effect; given it and age, there is slight evidence of a maternal smoking effect (LR statistic 3.55, df $= 1$, $P = 0.06$). The model fits well ($G^2 = 3.1$, df $= 8$).